Chaima Jaziri

Senior AI Engineer
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Summary

Applied AI Engineer specializing in production LLM and agentic systems for complex enterprise workflows. Experienced in taking AI products end-to-end from ambiguous user requirements and rapid prototyping through agent design, evaluation, enterprise integration, deployment, and production iteration. Strong focus on agent reliability, human-in-the-loop workflows, evaluation, context and tool-use strategies, and turning real-world failures and user feedback into measurable product improvements.

Skills

- **Agentic AI & GenAI** LangGraph, MCP, A2A, Agent Development Kit (ADK), RAG, Multi-Agent Systems, Tool Calling, Structured Outputs, Human-in-the-Loop, Context Engineering, Prompt Engineering, Guardrails.
- **Programming & Product Engineering** Python, TypeScript, Java, Bash, SQL, FastAPI, REST APIs, Microservices, Distributed Systems.
- **AI Evaluation & Reliability** LLM Evaluation, Agent Evaluation, Failure Analysis, Benchmarking, Regression Testing, LLM Observability, Feedback Loops, MLflow, LangSmith.
- **Systems & Integration:** REST APIs, FastAPI, Microservices, OAuth 2.0, OIDC, Azure AD, API Integration, Distributed Systems
- **Cloud & MLOps:** Azure, AWS, Docker, Kubernetes, GitHub Actions, MLflow, Azure ML, AWS SageMaker
- **Data & Infrastructure:** PostgreSQL, Redis, Pinecone, Azure AI Search, Spark, Kafka, Airflow, AWS S3
- **Observability:** Prometheus, Grafana, Datadog
- **Languages:** English (C1), French (C1)

Experience

Senior Applied AI Engineer | Ampliwork (Montreal) **2026 – Present**

- Lead the end-to-end development of production agentic AI experiences for a Fortune 500 enterprise, translating ambiguous business workflows into reliable AI-assisted products and owning delivery from discovery and rapid prototyping through evaluation, integration, deployment, and production iteration.
- Architect and build AI-powered document Cataloging and Transformation Mapping Agents that interpret enterprise documents, plan multi-step tasks, interact with tools and APIs, incorporate human review, and produce structured outputs inside existing customer workflows.
- Design agent behavior across prompting, context construction, tool use, orchestration, structured outputs, and human-in-the-loop interactions, balancing model autonomy with constrained workflows where reliability and user trust require stronger controls.
- Build evaluation and benchmarking pipelines using real customer workflows to measure task-level accuracy and diagnose failure modes; use findings to improve prompts, tools, orchestration, and evaluation methodology, increasing mapping accuracy from approximately 79% to 94%.
- Turn production failures and customer feedback into durable product improvements, identifying issues including unsupported document layouts, duplicate output generation, numerical evaluation mismatches, and workflow edge cases and translating them into fixes, regression scenarios, and product requirements.
- Work directly with client engineering, Product, DevOps, and Security teams to troubleshoot production issues, navigate ambiguous requirements, and rapidly ship solutions from proof of concept to production, while documenting architectures and operational practices for long-term ownership.

AI Engineer Intern | Ericsson / Aduna Global (Montreal) **May 2025 – May 2026**

- Led the design and delivery of agentic AI solutions for enterprise workflows, partnering with engineering teams to identify automation opportunities and turn them into production-oriented AI systems using LangGraph and LangSmith.
- Designed multi-agent architectures and defined agent responsibilities, tools, context strategies, evaluation methods, and integration patterns for reliable multi-step workflows.
- Built evaluation and feedback workflows to inspect agent behavior, diagnose failures, and iteratively improve prompts, orchestration, and tool-use strategies.
- Delivered a production PoC that reduced task turnaround time by 40% and manual intervention by 30%.
- Built LoRA fine-tuning pipelines on Azure ML with automated experimentation, deployment, and monitoring.
- Created onboarding documentation, internal playbooks, and technical workshops on Agentic AI, LangSmith evaluation, and responsible AI practices, accelerating adoption across engineering teams.
- Acted as a technical advisor on AI implementation decisions, helping engineers understand modern AI architectures and best practices.

Full-Stack AI Engineer | Beyond Camino (Remote) **Nov 2024 – Aug 2026**

- Designed and built an end-to-end AI product combining user-facing web experiences with intelligent AI assistants and retrieval-augmented generation.
- Built RAG and context retrieval pipelines supporting conversational workflows and knowledge-grounded AI responses.
- Improved AI responsiveness through caching, asynchronous processing, and streaming, reducing latency by 35% and improving the interactive user experience.
- Developed containerized deployment pipelines and reproducible environments supporting reliable production releases.
- Worked with stakeholders to translate product requirements and user needs into scalable AI-powered features.

AI Engineer | SBAT (Montreal) **Feb 2023 – June 2024**

- Developed production NLP systems that increased document classification accuracy from 74% to 91%, reducing manual processing.
- Built reproducible evaluation and experimentation workflows with MLflow across more than 30 model iterations.
- Productionized machine learning services using FastAPI, CI/CD, and monitoring.
- Worked with stakeholders to investigate model failures, improve data quality, and maintain reliable behavior across releases.

MLOps Engineer | Whitecape Technologie (Tunisia) **Jan 2022 – Mar 2023**

- Built and operated production machine learning infrastructure supporting SaaS recommendation systems.
- Designed scalable ML pipelines, automated retraining workflows, and production deployment strategies.
- Established CI/CD and infrastructure practices to improve release reliability.
- Introduced observability, monitoring, and data-drift detection to identify production degradation and improve long-term system reliability.

Education

Concordia University, Canada | MASc, Software Engineering **CGPA: 4.23/4.3**
Coursework: Generative AI, Deployment of AI Systems, Software Engineering, Social Ethics of AI.

National Engineering School of Sousse, Tunisia | Engineering Diploma, Appl. Comp. Sc. CGPA: 16.83/20
Coursework: Data Structures and Algorithms, Database Systems, Discrete Mathematics, Probability and Statistics

Research Experience

Thesis Project: Co-DynaDev, a Dynamic Multi-Agent Assistant for Automated Software Development

- Designed and implemented a multi-agent development assistant using LangGraph, FastAPI, Docker, and human-in-the-loop workflows for planning, execution, review, and iterative task completion.
- Researched memory architectures and context-management strategies that improved task completion by 31%, reduced hallucinations by 27%, and achieved over 99% passing tests.
- Evaluated prompting, memory, context, and tooling strategies using reproducible MLflow experiments to understand their effect on multi-step agent reliability.
- Designed workflows that allow users to review and steer agent execution rather than relying on unconstrained autonomous behavior.

Guardian MCP Guardrails

- Architected an enterprise-grade MCP security gateway enabling safe adoption of AI assistants across enterprise environments.
- Designed multi-layer safety validation combining fine-tuned LLMs, rule-based verification, and response filtering.
- Developed enterprise governance capabilities including DLP, ABAC authorization, anomaly detection, and policy enforcement.
- Integrated the platform with Claude Desktop and GitHub Copilot through MCP adapters, demonstrating secure enterprise AI deployment patterns.

Licenses & Certifications

- **Microsoft Certified: Azure AI Engineer Associate (AI-102)**
  Design, build, and deploy AI solutions on Azure for LLM integration, computer vision, and cloud-native AI applications.
- **IBM Machine Learning Professional Certificate**
  End-to-end machine learning curriculum covering model development, deployment, and production systems.
- **AWS Certified Generative AI Developer Professional** (In progress)
  Focused on enterprise Generative AI, Amazon Bedrock, Retrieval-Augmented Generation, and production AI architectures.